30/10/2026.

while loop :- A while loop is used to execute a block of code repeatedly as long as the given condition is True.

Syntax :

```
while condition :
    statements
    Increment/decreament.
```

Ex :-
```
i = 1
while i <= 5 :
    print(i)
    i = i+1
```

Nested loops :

Nested loops means writing one loop inside another loop.

In python, we can write :

- for loop inside for loop.
- while loop inside while loop
- for loop inside while loop
- while loop inside for loop.

→ for loop inside for loop

Syntax :

```
for itr_var1 in range() :
    for itr_var2 in range() :
        statements of inner loop
    Statements of outer loop.
```

Ex :-
```
for i in range(1, 4) :
    for j in range(1, 4) :
        print("*", end = " ")
    print()
```

o/p.

```
* * *
* * *
* * *
```